Analyzing Internal Representation Trajectories in GENCO

Cohort design, trajectory metrics, and findings

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Contents

Summary 3

1. Research motivation 3
1.1 Questions motivating the study . . . 3
How does GENCO revise its internal representation? . . . 3
What distinguishes difficult cases? . . . 3
Which entities account for the movement? . . . 4
Does trajectory geometry add information beyond final metrics? . . . 4
Which questions remain open? . . . 4

2. What the trajectory represents 4

3. Designing a contrasting scenario cohort 5
3.1 Familiarity from the nearest training reference . . . 5
3.2 Physical quality from mean absolute PBE . . . 5
3.3 Why use percentiles? . . . 5
3.4 The four contrasting groups . . . 6

4. Selecting representative scenarios 7
4.1 Corner score . . . 7
4.2 Inner-boundary score . . . 7
4.3 Physical-diversity representative . . . 8
4.4 Final cohort . . . 8

5. Trajectory metric definitions 8
5.1 Step movement . . . 9
5.2 Path length . . . 9
5.3 Net displacement . . . 9
5.4 Path efficiency . . . 9
5.5 Local directional consistency . . . 9
5.6 Global directional alignment . . . 9
5.7 Timing and magnitude summaries . . . 10

6. Aggregation strategy 10

7. Descriptive results 10
7.1 Physical quality has the stronger relationship with trajectory geometry . . . 10
7.2 Familiarity has a similar but weaker relationship . . . 11
7.3 The four groups follow a coherent ordering . . . 11
7.4 Rank relationships support the group contrasts . . . 12
7.5 Movement is concentrated in the early transitions . . . 12
7.6 Bus and generator trajectories differ . . . 13
7.7 Movement magnitude and directness are distinct . . . 13
7.8 Movement is bus-dominated but distributed . . . 13
7.9 Latent-movement hotspots do not identify final PBE hotspots . . . 14

8. Interpretation 14

9. Scope of the findings 14

Summary

Final embeddings describe where a model's internal representation ends, but they do not describe how it arrived there. Two GENCO predictions can have similar final errors or similar final embeddings while following different internal paths across the model. One representation may change substantially in the early layers and then stabilize. Another may continue to turn, reverse, or make large updates later in the network.

This study examines layer-wise latent representations for a deliberately selected cohort of 20 held-out IEEE 14-bus scenarios. The cohort contrasts two properties measured at the final model output:

1. familiarity, defined by distance to the nearest training scenario in pooled final-embedding space; and
2. physical prediction quality, defined by mean absolute power-balance error.

Five scenarios were selected from each of four groups: familiar-good, familiar-bad, unfamiliar-good, and unfamiliar-bad. For every selected scenario, we captured 12 prediction-head-input latent embeddings for each bus and generator and derived geometric trajectory measures across the 11 layer transitions.

The principal descriptive result is that physically bad scenarios exhibit longer, less efficient, and less directionally consistent latent trajectories than physically good scenarios. Unfamiliar scenarios show the same pattern, but the differences are smaller. Most latent movement occurs in the first five transitions, while the bad groups reach their largest average movement later than the good groups. These findings describe the selected cohort and do not establish causal relationships or population-level performance.

1. Research motivation

GENCO produces physical predictions through a sequence of learned message-passing and latent-update operations. Standard evaluation focuses on the final output, such as voltage predictions, generator predictions, constraint violations, or power-balance error. Final-embedding analysis adds information about where the learned representation lies relative to training scenarios, but it still treats inference as a single endpoint.

The internal trajectory provides a different object of study. For an entity e, let

$$z_e^{(1)}, z_e^{(2)}, \ldots, z_e^{(12)}$$

denote the 12 captured latent representations across the model layers. The trajectory records how the internal state of a bus or generator changes before the final prediction is produced.

The motivation is not that movement is inherently good or bad. The aim is to determine whether measurable properties of the path provide information that is absent from the endpoint alone. In particular, the study asks whether difficult scenarios are associated with more movement, less direct movement, later revisions, or different localization patterns.

1.1 Questions motivating the study

The analysis was designed around five groups of questions.

How does GENCO revise its internal representation?

- How much latent movement occurs across the model?
- Is movement concentrated in the early layers or distributed across the network?
- At which transition does each entity move most?
- Do trajectories proceed in a consistent direction, or do they turn and reverse?

What distinguishes difficult cases?

- Do scenarios with high final physical error have longer latent paths?
- Do scenarios far from the training-reference neighborhood behave differently?

- Does physical quality have a stronger relationship with trajectory geometry than familiarity?
- Can scenarios with similar endpoints have different internal paths?

Which entities account for the movement?

- Do buses and generators follow similar latent trajectories?
- Which entities contribute most to the total movement within a scenario?
- Are the largest-moving entities also the locations of the largest final power-balance errors?
- Is movement concentrated in one entity or distributed across the network?

Does trajectory geometry add information beyond final metrics?

- Are path length and final displacement measuring different properties?
- Does path directness separate cases with similar movement magnitude?
- Does the timing of movement distinguish good and bad scenarios?

Which questions remain open?

The original motivation also included step-level physical improvement and changes in the nearest-reference neighborhood. Those questions require decoded physical predictions or pooled scenario embeddings at every layer. The current capture contains entity-level latent embeddings at each layer, but not step-level decoded power-balance error or step-level nearest-reference retrieval. The present report therefore does not claim to measure physical convergence across layers.

2. What the trajectory represents

The captured states are the 384-dimensional bus and generator tensors supplied to their respective prediction heads at each of the 12 model layers. The most precise description is:

> Layer-wise prediction-head-input latent embeddings.

These states are internal learned representations. They are not voltages, generator dispatch values, physical residuals, confidence scores, or classical solver iterates.

The field named `correction_step` in the trajectory table indexes model layers from 1 through 12. A transition from step 4 to step 5 is therefore a transition between layer-wise latent states. It can reflect message passing, latent updates, and the model's physics-informed computations. It should not be interpreted as one isolated physical correction or one conventional solver iteration.

The capture contains:

Quantity	Count
Selected scenarios	20
Bus trajectories	280
Generator trajectories	85
Total entity trajectories	365
Snapshots per entity	12
Layer transitions per entity	11
Total trajectory rows	4,380
Total transition rows	4,015
Embedding dimension	384

Every scenario contains 14 buses and either four or five generators. The final captured state corresponds to the native final embedding exported by GENCO, which anchors the internal trajectory to the final representation used in the familiarity analysis.

3. Designing a contrasting scenario cohort

Capturing layer-wise entity embeddings for all 15,000 test scenarios would create a much larger artifact before the semantics and usefulness of the metrics were understood. The initial study therefore uses a small, deliberately contrasting cohort.

The cohort is not a random sample. It is an experimental design intended to provide clear contrasts along two final-output axes:

1. distance from the training-reference representation; and
2. final physical prediction error.

3.1 Familiarity from the nearest training reference

For each held-out test scenario s, bus embeddings are averaged, generator embeddings are averaged, and the two pooled vectors are concatenated into one scenario representation z_s.

Familiarity is measured by the Euclidean distance to the nearest pooled training representation:

$$d(s) = \min_{r \in \mathcal{R}_{\text{train}}} \|z_s - z_r\|_2$$

A lower value means that the final test representation is closer to at least one training scenario under this pooling and distance policy. It does not mean that the two scenarios are physically identical or that the prediction is reliable.

The nearest-reference distances are percentile-ranked across all 15,000 test scenarios. The distance percentile describes the relative position of a test scenario among other test scenarios' nearest-training distances. It is not a percentile over all individual training distances.

3.2 Physical quality from mean absolute PBE

Physical quality is measured using the scenario's mean absolute bus power-balance error:

$$r(s) = \frac{1}{N_{\text{bus}}} \sum_{i=1}^{N_{\text{bus}}} |\text{PBE}_i(s)|$$

These residual values are also percentile-ranked across the 15,000 test scenarios. A low residual percentile indicates relatively low final physical error, while a high percentile indicates relatively high error.

One residual definition is used consistently throughout cohort selection. Maximum PBE is retained for localization and physical-diversity selection, but it does not define the good and bad groups.

3.3 Why use percentiles?

Raw embedding distance and PBE have different units, scales, and distribution shapes. Fixed raw thresholds would be specific to this model, dataset, and representation policy. Min-max normalization would also be sensitive to a small number of extreme values.

Empirical percentile ranks provide a common two-dimensional coordinate system:

$$\big(d_p(s), r_p(s)\big)$$

where $d_p(s)$ is the nearest-training-distance percentile and $r_p(s)$ is the mean-absolute-PBE percentile.

The 25th and 75th percentiles define low and high regions:

```
familiar    = distance percentile <= 0.25
unfamiliar  = distance percentile >= 0.75
good        = residual percentile <= 0.25
bad         = residual percentile >= 0.75
```

The middle 50 percent on either axis is not used for this contrasting cohort because those scenarios provide weaker separation.

The corresponding raw quartile thresholds were:

Quantity	25th percentile	75th percentile
Nearest-training distance	0.129231	0.170135
Mean absolute PBE	0.000583	0.001211

3.4 The four contrasting groups

The percentile rules define four groups:

Group	Distance percentile	Residual percentile	Interpretation
Familiar-good	≤ 0.25	≤ 0.25	Near a training representation and physically good
Familiar-bad	≤ 0.25	≥ 0.75	Near a training representation but physically poor
Unfamiliar-good	≥ 0.75	≤ 0.25	Far from training representations but physically good
Unfamiliar-bad	≥ 0.75	≥ 0.75	Far from training representations and physically poor

The mixed groups are particularly informative. Familiar-bad scenarios challenge the assumption that proximity to a training representation implies a good physical result. Unfamiliar-good scenarios show that distance from the training-reference neighborhood does not necessarily imply failure.

Across all test scenarios, nearest-training distance and mean absolute PBE have a Spearman correlation of 0.494. The association is positive but incomplete, leaving meaningful mixed cases for study.

The candidate counts were:

Group	Eligible test scenarios
Familiar-good	1,567
Familiar-bad	275
Unfamiliar-good	218
Unfamiliar-bad	2,110

In total, 4,170 scenarios fall into one of the four corner regions and 10,830 lie outside them. The asymmetry in group sizes is itself informative. Cases that are familiar but physically bad, or unfamiliar but physically good, are less common than the aligned groups, but they are not absent.

4. Selecting representative scenarios

Randomly selecting five scenarios from each group would make the study difficult to reproduce and could yield weak contrasts. Selecting only the most extreme scenarios would create a different problem by overrepresenting unusual outliers. The final design combines extreme representatives, near-threshold representatives, and one physically diverse case.

Each group contains:

```
3 corner representatives
1 inner-boundary representative
1 physical-diversity representative
```

4.1 Corner score

Each group has an ideal corner in percentile space:

Group	Corner
Familiar-good	$(0, 0)$
Familiar-bad	$(0, 1)$
Unfamiliar-good	$(1, 0)$
Unfamiliar-bad	$(1, 1)$

For scenario s, the corner score is:

$$C(s) = \sqrt{\left(d_p(s) - d_c\right)^2 + \left(r_p(s) - r_c\right)^2}$$

where (d_c, r_c) is the group's ideal corner. A smaller score means the scenario is a stronger extreme representative of the group.

Eligibility is determined before corner ranking. A scenario in the middle of the distribution cannot enter a group merely because it lies close to one coordinate of the corner.

For example, familiar-bad scenario `215418` has:

```
distance percentile = 0.02433
residual percentile = 0.95087
corner              = (0, 1)
corner score        = 0.05483
```

It is close to the familiar-bad extreme because its representation is among the most training-like while its final mean absolute PBE is among the highest.

4.2 Inner-boundary score

The inner-boundary point represents the minimum percentile conditions for clear membership in a group:

Group	Inner boundary
Familiar-good	$(0.25, 0.25)$
Familiar-bad	$(0.25, 0.75)$
Unfamiliar-good	$(0.75, 0.25)$
Unfamiliar-bad	$(0.75, 0.75)$

For scenario s, the boundary score is:

$$B(s) = \sqrt{\left(d_p(s) - d_b\right)^2 + \left(r_p(s) - r_b\right)^2}$$

where (d_b, r_b) is the group's inner-boundary point.

A low boundary score identifies a scenario that clearly satisfies the group definition but is less extreme. This reduces dependence on a small number of corner outliers.

For example, familiar-bad scenario `86876` has:

```
distance percentile = 0.23277
residual percentile = 0.75860
inner boundary      = (0.25, 0.75)
boundary score      = 0.01926
```

It is familiar and bad under the quartile definitions, but it lies close to the threshold intersection rather than the extreme corner.

4.3 Physical-diversity representative

After selecting three corner cases and one boundary case, the final scenario in each group is chosen to add physical diversity.

The selection prioritizes:

1. a maximum-PBE bus not already represented in the group;
2. a nearest training reference not already represented;
3. separation in maximum-PBE percentile from the selected cases;
4. corner score; and
5. scenario identifier as the deterministic final tie-breaker.

This policy prevents the five scenarios in one group from differing only in percentile rank while sharing the same residual hotspot or nearest reference.

4.4 Final cohort

The selected cohort is:

Group	Corner representatives	Inner boundary	Physical diversity
Familiar-good	189231, 212542, 28220	14412	89840
Familiar-bad	215418, 16214, 239841	86876	261589
Unfamiliar-good	180647, 94412, 124340	159813	18917
Unfamiliar-bad	31048, 71385, 58428	177691	103415

The design deliberately balances strong contrasts with less extreme and physically distinct cases. It is suitable for a descriptive trajectory study, but it is not a representative sample of the full test distribution.

5. Trajectory metric definitions

For entity e, define the movement vector between consecutive layers as:

$$\Delta_e^{(t)} = z_e^{(t+1)} - z_e^{(t)}, \qquad t = 1, \ldots, 11$$

The metrics are computed independently for every bus and generator trajectory.

5.1 Step movement

$$m_e^{(t)} = \left\| \Delta_e^{(t)} \right\|_2$$

Step movement measures the size of one layer-to-layer latent update. It supports transition profiles and identification of the largest movement step.

5.2 Path length

$$L_e = \sum_{t=1}^{11} \left\| \Delta_e^{(t)} \right\|_2$$

Path length measures the total amount of latent movement accumulated across the trajectory. It depends on every intermediate state.

5.3 Net displacement

$$D_e = \left\| z_e^{(12)} - z_e^{(1)} \right\|_2$$

Net displacement measures the straight-line distance between the first and final captured representations. It ignores the route between them.

5.4 Path efficiency

$$E_e = \frac{D_e}{L_e}$$

For a nonzero path, efficiency lies between 0 and 1. A value near 1 indicates that most movement follows a direct route from the first to final state. A lower value indicates turning, reversal, or indirect movement.

Path efficiency is a geometric property of latent space. It is not physical efficiency and does not prove convergence.

5.5 Local directional consistency

$$K_e^{\text{local}} = \frac{1}{10} \sum_{t=1}^{10} \cos\left(\Delta_e^{(t)}, \Delta_e^{(t+1)} \right)$$

This metric compares each movement vector with the next movement vector.

- Values near 1 indicate continued movement in a similar direction.
- Values near 0 indicate approximately orthogonal successive movements.
- Negative values indicate directional reversal.

The trajectory-level value is the mean over valid consecutive pairs.

5.6 Global directional alignment

Let

$$g_e = z_e^{(12)} - z_e^{(1)}$$

be the overall displacement vector. Global alignment is:

$$K_e^{\text{global}} = \frac{1}{11} \sum_{t=1}^{11} \cos\left(\Delta_e^{(t)}, g_e\right)$$

This metric measures whether each transition is aligned with the trajectory's eventual first-to-final direction.

Local consistency and global alignment answer different questions. A trajectory can make locally consistent movements while curving away from its overall displacement, or alternate locally while still producing a meaningful net change.

5.7 Timing and magnitude summaries

For every entity, the analysis also records:

- mean step movement;
- maximum step movement;
- the starting layer of the maximum movement transition;
- path-length rank within the scenario;
- path-length rank within the entity type;
- share of total scenario path length; and
- share of bus-only or generator-only path length.

These measures support temporal profiling and entity localization.

6. Aggregation strategy

Trajectory metrics are first computed at entity level. Scenario summaries are then calculated separately for:

- buses;
- generators; and
- all entities.

Each scenario summary includes total path length, total displacement, distributional summaries of entity metrics, and the highest-movement entity.

Group comparisons are scenario-balanced. Each scenario contributes one summary to its cohort group, regardless of whether it contains four or five generators. This prevents scenarios with more generators from receiving greater weight in the group comparison.

Total path length and mean entity path length should not be conflated. Total path length measures aggregate activity and depends partly on entity count. Mean entity path length is safer when comparing typical per-entity movement across scenarios with different generator counts.

7. Descriptive results

7.1 Physical quality has the stronger relationship with trajectory geometry

Combining the two good groups and comparing them with the two bad groups gives:

Metric	Good	Bad	Relative difference
Mean total path length	161.94	211.11	+30.36%
Mean entity path length	8.95	11.48	+28.28%
Mean net displacement	5.64	6.85	+21.45%
Path efficiency	0.6165	0.5908	-4.18%
Local directional consistency	0.4727	0.4251	-10.07%
Global directional alignment	0.4517	0.4193	-7.17%

Within the selected cohort, physically bad scenarios exhibit more latent movement, larger first-to-final changes, lower path efficiency, and weaker directional consistency.

The relationship is visible within both familiarity categories:

Contrast	Total path increase for bad cases	Change in local consistency
Familiar-bad versus familiar-good	+33.68%	-12.37%
Unfamiliar-bad versus unfamiliar-good	+27.47%	-7.60%

This indicates that the good-bad distinction is not solely produced by the familiarity grouping.

7.2 Familiarity has a similar but weaker relationship

Combining both familiar groups and comparing them with both unfamiliar groups gives:

Metric	Familiar	Unfamiliar	Relative difference
Mean total path length	175.93	197.12	+12.04%
Mean entity path length	9.67	10.76	+11.17%
Mean net displacement	5.99	6.50	+8.55%
Path efficiency	0.6115	0.5958	-2.55%
Local directional consistency	0.4586	0.4392	-4.23%
Global directional alignment	0.4421	0.4289	-2.98%

Unfamiliar scenarios tend to move farther and less directly, but the differences are smaller than those associated with final physical quality.

7.3 The four groups follow a coherent ordering

Scenario-balanced combined-entity results are:

Group	Total path	Mean entity path	Net displacement	Efficiency	Local consistency	Global alignment
Familiar-good	150.57	8.37	5.39	0.6262	0.4888	0.4643
Unfamiliar-good	173.31	9.53	5.89	0.6069	0.4565	0.4391
Familiar-bad	201.29	10.98	6.58	0.5967	0.4284	0.4199
Unfamiliar-bad	220.92	11.98	7.11	0.5848	0.4219	0.4187

The ordering is approximately:

```
familiar-good
    -> unfamiliar-good
    -> familiar-bad
    -> unfamiliar-bad
```

Moving along this ordering corresponds to increasing path length and displacement, together with decreasing

efficiency and directional consistency.

7.4 Rank relationships support the group contrasts

Across the 20 selected scenario summaries, the descriptive Spearman correlations are:

Trajectory metric	Training distance	Mean absolute PBE	Maximum absolute PBE
Total path length	+0.462	+0.753	+0.716
Mean entity path length	+0.459	+0.726	+0.678
Mean net displacement	+0.418	+0.732	+0.687
Path efficiency	-0.522	-0.603	-0.602
Local consistency	-0.445	-0.719	-0.710
Global alignment	-0.383	-0.624	-0.615

Higher PBE is associated with longer movement and lower directional consistency. Nearest-training distance has the same directional relationships but more moderate magnitudes.

These correlations are descriptive values for an intentionally selected cohort. They are not significance-tested population estimates.

7.5 Movement is concentrated in the early transitions

The fraction of mean path movement occurring during transitions 1 through 5 is:

Group	Share in transitions 1 through 5
Familiar-good	83.23%
Unfamiliar-good	85.84%
Familiar-bad	87.57%
Unfamiliar-bad	87.18%

Only a small fraction occurs during transitions 9 through 11:

Group	Share in transitions 9 through 11
Familiar-good	6.21%
Unfamiliar-good	4.92%
Familiar-bad	4.17%
Unfamiliar-bad	3.95%

The latent process is therefore strongly front-loaded. Later layers continue to change the representations, but their average contribution to total path length is much smaller.

The transition with the largest average movement differs by physical-quality group:

Group	Largest average transition	Mean movement
Familiar-good	2 to 3	1.958
Unfamiliar-good	2 to 3	2.012
Familiar-bad	4 to 5	2.377
Unfamiliar-bad	4 to 5	2.631

The good groups reach their largest average movement earlier, while both bad groups peak at transition 4 to 5. This is evidence of a later latent reconfiguration, not evidence that physical convergence occurs at that layer.

7.6 Bus and generator trajectories differ

Across all selected scenarios:

Metric	Buses	Generators
Entity count	280	85
Mean path length	10.939	7.854
Mean net displacement	6.733	4.656
Mean efficiency	0.616	0.562
Mean local consistency	0.554	0.101
Mean global alignment	0.487	0.264

Bus representations move farther and follow substantially more consistent directions. Generator representations travel shorter paths but show much lower local consistency and global alignment.

These values should not be interpreted as evidence that generators are physically unstable. Bus and generator states are produced through different learned pathways and prediction heads. Their latent geometry has no direct physical unit.

7.7 Movement magnitude and directness are distinct

Scenario `71385` has the largest total path length:

```
group: unfamiliar-bad
total path length: 281.56
mean entity path: 14.82
mean net displacement: 9.38
mean efficiency: 0.617
```

Scenario `58428` has a shorter total path of `237.15` but a lower mean efficiency of `0.554`.

The comparison shows why both quantities are needed:

- path length measures how much movement occurred;
- efficiency measures how directly the trajectory reached its endpoint.

A long path is not necessarily a strongly meandering path.

The familiar-good group is also more uniform. Its five total path lengths range from `148.77` to `153.43`, with a standard deviation of `1.57`. The familiar-bad and unfamiliar-bad groups have standard deviations of `26.01` and `40.20`, respectively.

7.8 Movement is bus-dominated but distributed

The highest-movement entity is a bus in all 20 selected scenarios. Bus `3` is the highest-movement entity in 11 scenarios, and bus `5` is highest in seven.

The largest individual path occurs for bus `3` in scenario `58428`:

```
path length: 22.956
share of scenario path: 9.68%
```

Even the largest-moving entity contributes only about 7 to 10 percent of its scenario's total path length. The movement is therefore localized enough to identify repeated hotspots but remains distributed across many entities.

7.9 Latent-movement hotspots do not identify final PBE hotspots

The highest-movement bus matches the maximum-PBE bus in only two of the 20 scenarios. The maximum-PBE bus appears among the top three movement ranks in four scenarios. Its median movement rank is 5, and its mean rank is 5.55.

This negative result is important. The bus whose representation moves most is usually not the bus with the largest final physical balance error. Latent movement likely reflects distributed message passing and computation rather than a direct marker of final error location.

Movement should therefore not be used as a proxy for physical-error localization without additional evidence.

8. Interpretation

The selected cohort supports three main interpretations.

First, final physical quality has a stronger relationship with latent trajectory geometry than final familiarity does. Bad scenarios exhibit more movement and less direct, less directionally consistent paths within both familiar and unfamiliar categories.

Second, the trajectory contains temporal information that is absent from the final embedding. Good and bad scenarios differ not only in total movement but also in when the largest average movement occurs.

Third, latent activity is distributed across entities. Repeated bus hotspots exist, but they do not generally coincide with final PBE hotspots. The trajectory therefore appears to characterize distributed internal computation rather than simply reproducing the spatial pattern of the final residual.

These interpretations remain geometric and descriptive. They do not establish that long or indirect paths cause physical error, that the model is reasoning in a human-like sense, or that latent movement corresponds to a classical optimization trajectory.

9. Scope of the findings

- The cohort contains five selected scenarios per group, not a random or population-representative sample.
- Selection uses final distance and final PBE, so the cohort is designed to amplify contrasts along those variables.
- Correlations are descriptive and are not accompanied by significance tests or uncertainty intervals.
- The captured states are layer-wise prediction-head inputs, not decoded physical states.
- No PBE or constraint trajectory was captured at each layer.
- No changing nearest-reference neighborhood was computed across layers.
- Bus and generator latent metrics are not directly comparable to physical quantities.
- The results apply to the IEEE 14-bus seed-42 model and should not be assumed to generalize to other grids, checkpoints, or model architectures.